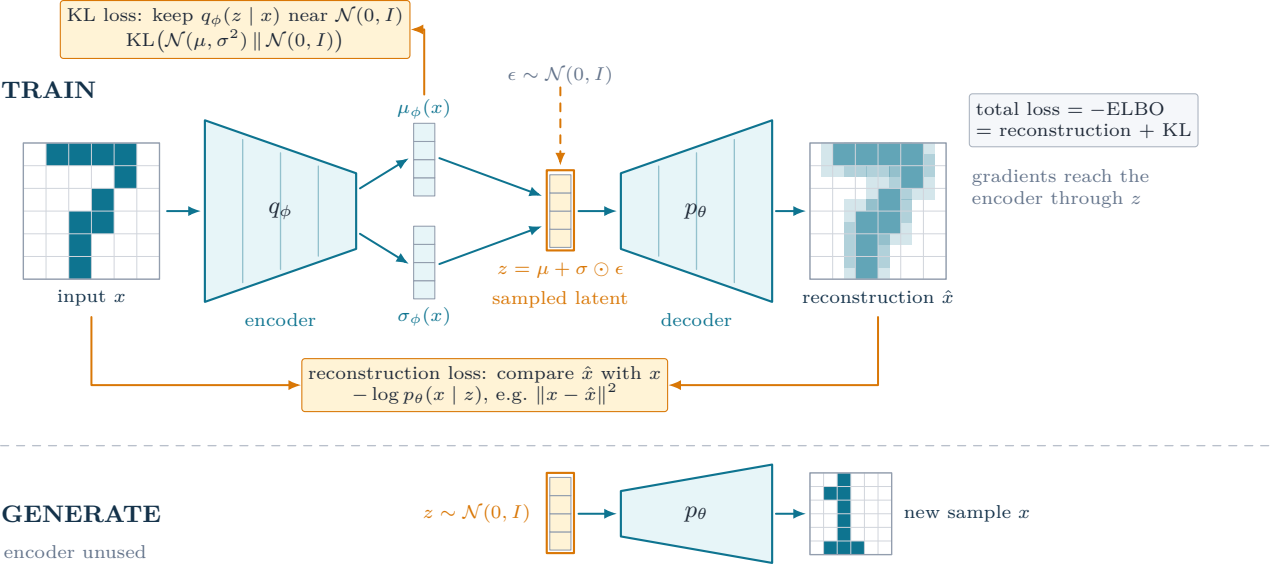
KL loss: keep $q_\phi(z \mid x)$ near $\mathcal{N}(0, I)$
$\mathrm{KL}\big(\mathcal{N}(\mu, \sigma^2) \,\|\, \mathcal{N}(0, I)\big)$
TRAIN
$\epsilon \sim \mathcal{N}(0, I)$
$\mu_\phi(x)$
q_ϕ
p_θ
total loss = −ELBO
= reconstruction + KL
gradients reach the
encoder through z
$z = \mu + \sigma \odot \epsilon$
sampled latent
input x
encoder
$\sigma_\phi(x)$
decoder
reconstruction $\hat{x}$
reconstruction loss: compare $\hat{x}$ with x
$-\log p_\theta(x \mid z)$, e.g. $\|x - \hat{x}\|^2$
GENERATE
encoder unused
$z \sim \mathcal{N}(0, I)$
p_θ
new sample x